

Robust Fine-Tuning of Deep Neural Nets with Hessian-based Generalization Guarantees

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PROBLEM STATEMENT

Fine-tuning has been a modern approach for transfer learning: Download a pretrained network; Fine-tune it on a target task.

What is the generalization error of a fine-tuned model?

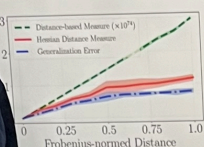
Let f_w be an L -layer feed-forward neural network initialized with pretrained weights $W^{(0)}$. Let $\ell(f_w(x), y)$ be the loss of f_w bounded between 0 and 1 on input sample (x, y) . **Problem statement:** How large is the generalization error of a fine-tuned neural network f_w :

$$L(f_w) - \hat{L}(f_w),$$

where $L(f_w)$ is the population loss and $\hat{L}(f_w)$ is the empirical loss.

BACKGROUND AND RELATED WORKS

- Distance from initialization is identified as the key generalization property of fine-tuned models in previous works [GHP'21; LZ'21].
- Our observation: In addition to distance from initialization, **Hessians also affect generalization**.



Bounds for deep neural networks:

Analysis is an approach for deriving generalization bounds for deep learning models [McA'99; McA'01].

Studying generalization errors of deep neural nets [SK18; NBS18].

to different black-box models, including graph neural networks [LZ21] and data augmentation [CFD'21].

(GN+18) is a key notion to quantify the loss stability of perturbations.

$$\sigma(f_w) = \mathbb{E}[\ell(f_w(x), y)] - \ell(f_w(x), y)$$

random noise U .

HESSIAN-BASED GENERALIZATION BOUNDS

Claim. The noise stability can be approximated by the matrix product between the covariance matrix of the noise and Hessians of the loss:

$$I(f_w) = \sum_{i=1}^L \Sigma_i H_i[\ell(f_w(x), y)] + \xi$$

- Proof.** Consider Taylor's expansion on the noise stability $I(f_w)$.

σ	ResNet-18		ResNet-50	
	Noise Stability	Hessian Approx.	Noise Stability	Hessian Approx.
0.011	1.13 ± 0.24	1.29	1.12 ± 0.20	1.30
0.012	1.45 ± 0.29	1.44	1.56 ± 0.26	1.79
0.013	1.82 ± 0.35	1.80	2.10 ± 0.33	2.10
0.014	2.23 ± 0.40	2.09	2.71 ± 0.38	2.43
0.015	2.63 ± 0.45	2.34	3.30 ± 0.40	2.79
Relative BSS	0.75%		2.98%	

Main result. The generalization error of a fine-tuned model is characterized by the Hessian matrix of model weights and the weight vector subtracted from initialization across all layers.

- $H_i^*[\ell(f_w(x), y)]$: Hessian of the loss at given example (x, y) w.r.t. layer i .
- w_i : The flattened weight vector $W_i - W_i^{(0)}$ at layer i .
- The generalization error of a fine-tuned model f_w scales as:

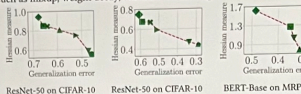
$$O\left(\frac{\sum_{i=1}^L \sqrt{w_i^T H_i^*[\ell(f_w(x), y)] w_i}}{\sqrt{n}}\right)$$

Results applicable to various fine-tuning methods.

- Distance-based regularization [GHP'21]
 - Consistent loss reweighting under class conditional label noise [NDR'13]
- The generalization bound scales with the product between the trace of Hessian and distance from initialization.

EVALUATION OF HESSIAN MEASURES

Evaluate the Hessian measure $\sum_i w_i^T H_i^*[\ell(f_w(x), y)] w_i$ for seven fine-tuning methods, such as mixup, weight decay, and label smoothing.



Numerically calculate the generalization bound using ResNet-50 on CIFAR datasets and BERT-Base on MRPC and SST2 datasets.

Generalization bound	CIFAR-10	CIFAR-100	MRPC	SST2
Arora et al. (2018)	1.62E+06	9.66E+07	/	/
Neyshabur et al. (2018)	3.18E+11	1.62E+13	/	/
Our result	2.26	7.23	3.83	9.71

We compute the Hessian vector product using PyHessian [YKG'20].

EXPERIMENTAL RESULTS

Algorithm. Our algorithm incorporates two components for fine-tuning with noisy labels:

- Distance-based regularization: Constrain the distance of weights from initialization $\|W_i - W_i^{(0)}\|_F$ [GHP'21].
- Consistent loss reweighting: Reweight loss with the inverse of label confusion matrix (parameterized by a single noise rate or estimated from model predictions) [PRK* '17; XLW* '19; YLH* '20].

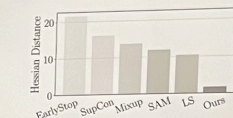
Fine-tuning under two label noise settings on six image classification tasks from DomainNet dataset.

- Independent label noise: Randomly flip the labels with a given noise rate.
- Correlated label noise: Use the predicted labels of models trained from other domains in DomainNet dataset [MCS+21].

Fine-tuning ResNet-18 on two label noise settings

Dataset	Independent label noise		Correlated label noise	
	Sketch (40%)	Sketch (60%)	Clipart (41.5%)	Sketch (47.7%)
Label smooth	74.69±1.97	55.35±1.60	55.35±1.60	64.90±2.93
SupCon	75.14±1.73	61.06±3.20	61.06±3.20	63.92±2.15
SAM	77.63±2.16	64.53±2.84	64.53±2.84	67.27±2.39
Ours	81.96±0.98	70.00±1.71	70.00±1.71	71.84±2.72

Key observation. Evaluate the Hessian measure $\sum_i w_i^T H_i^*[\ell(f_w(x), y)] w_i$ of models fine-tuned by different algorithms.



Extension to other settings.

- Fine-tuning RoBERTa-Base on MRPC dataset (text classification task) with independent label noise.
- Fine-tuning Vision Transformer models on image classification tasks from DomainNet dataset with correlated label noise.

CONCLUSION

- We develop Hessian-based generalization bounds for fine-tuned models.
- Hessian-based measures accurately correlate with generalization errors.
- Our analysis leads to an algorithm for fine-tuning with noisy labels.

Links:

- Paper: <https://arxiv.org/pdf/2206.02659.pdf>
- Code: <https://github.com/NEU-StatsML-Research/Robust-Fine-Tuning>